A Thirteen-Integer Jones Equation
Old-Source Comparison with CRT and Coprime Guards

Mathematical research artifact for ProveIt

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1 Thirteen integers: CRT elimination and a coprime spectator guard

The polynomial specified below has exactly thirteen unrestricted integer unknowns, exact total degree 28 847 344, and a division-free circuit of 754 live binary operations (516 multiplications, 215 additions and 23 subtractions). Its integer zero existence is equivalent to a finite syntactic ZFC refutation. Thus, with the stated inherited compiler theorem, ZFC proves that its lack of integer zeros is equivalent to Con(ZFC). This is an ordinary mathematical proof, not a new Lean certificate, a proof of consistency, or a global minimality claim. Unlike the ten-integer companion, the arithmetic finalizer here uses only the established Jones ratio interface, not Sun's 2026 bridge.

1.1 A coherent fixed source and an explicit polynomial

Let $p_1, \dots, p_{22}$ be the original package-31 quadratic residuals after substituting $x_i = n_i + 1$ in all 31 positive source coordinates. The exact source is `upstream/parent_quartic31.json`; the complete fixed port mapping is $(\beta, K_0, M_C, M_F) = (\beta, K, M_C, M_{F,\mathrm{source}})$ from `FIXED_BINDING_OLD_SOURCE.json`. All four compressed little-endian payloads are included and hash-authenticated. This uses the original transport coordinate, not package 45's extra transport shift. The imported source theorem says

$$\exists n \in \mathbb{N}^{31} \ \bigwedge_{j=1}^{22} p_j(n) = 0 \quad \Longleftrightarrow \quad \text{there is a finite ZFC refutation.}$$

The fifth residual equals 1 at $n = 0$, so any common zero is nonzero.

Use the selective weights w_i, row order, and targets q_j in `CODE.json`, identical to the companion's certified tuple. Their maxima are $W = 183$ and $Q_* = 5190$; put $K_* = Q_* + 2W = 5556$ and $E_* = K_* + W + 2 = 5741$. For $p_j(n) = \sum_\alpha p_{j,\alpha} n^\alpha$, set

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}, \qquad D_c(B) = \sum_{j,\alpha} c_{j,\alpha} B^{q_j - \alpha \cdot w}.$$

The finite isolation certificate checks all 71 808 possible interactions of the 136 used source terms with the 528 padded quadratic monomials. It proves

$$[t^{q_j}]\Big(1 + \sum_i n_i t^{w_i}\Big)^2 D_c(t) = 2p_j(n).$$

It also proves that the displayed exponents in D_c are distinct and its leading coefficient is 2. All $|c_{j,\alpha}| < A_* = 2^{2134927}$; the verifier derives this bound from the actual fixed-port bit lengths.

The unknowns are

$$(\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, r, v) \in \mathbb{Z}^{13}.$$

Every subsequent symbol is a polynomial register, not an additional unknown. Fix

$$\kappa = 2134940, \quad a = 2^{\kappa} = 8192A_*, \quad \eta = 2^{16012040} = 2^{15\kappa/2-10},$$

and define

$$\begin{aligned}
b &= 2\epsilon^2, & B &= ab^2,\\
\ell &= \sum_i B^{w_i}, & T_* &= \sum_j B^{q_j},\\
N_0 &= B^{W+1}, & L &= 2bB^{E_*},\\
N &= \eta B^{2966}, & c &= bB^{W},\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2(1+g)^2 D_c(B) + T_*,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)T_*,\\
R &= L^2 + (S_c + T_c)L + (L - 1 - S_c), & G_c &= 32c.
\end{aligned}$$

All powers have fixed nonnegative integer exponents. In particular, N is an ordinary polynomial, not a variable-exponent abbreviation.

Use the Jones registers with the exact CRT reparameterization of H below, and $\omega = w^2$:

$$\begin{aligned}
U &= N^2\omega, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= b\omega,\\
C &= B_1 + W_1 + \phi, & J &= 4A - 5, & Q &= 3W_1C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= C + F(B_1 + jC), & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
u &= J, & t &= Q.
\end{aligned}$$

Set

$$\delta = C - KY, \qquad \Delta = g(c-g)K^2 - c^2\delta^2, \qquad (d_1, d_2, d_3) = (3s-1, 3\phi+2, 3\Delta-1).$$

The new coprime-spectator stack is

$$C_0 = a_0, \qquad B_j^* = 1 + 4d_jC_{j-1}z_j^2, \qquad C_j = C_{j-1}B_j^* \quad (j = 1, 2),$$

$$B_3^* = 1 + 4d_3C_2a_1z_3^2, \qquad C_* = C_3 = C_2B_3^*.$$

Thus the last guard contains the second square candidate a_1 as a spectator. This is the only change to the Pell sign stack. Finally define

$$\begin{aligned}
Z &= \epsilon w i z_1 z_2 z_3 a_0, & \Sigma &= Zu^5, & \rho &= t + ur,\\
\Theta &= (\rho^3 - u^2\rho - u^3)(\rho^2 + 23u^2), & X &= \Sigma v - \Theta,\\
A_0 &= \Sigma^2 C_*, & A_1 &= \Sigma^2 a_1.
\end{aligned}$$

The requested thirteen-integer equation is the following single polynomial:

$$\boxed{\mathcal{F}_{13} = (X^2 - A_0 - A_1)^2 - 4A_0A_1 = 0.} \tag{1}$$

The literal division-free DAG is `generated/signed_coprime_crt.json`.

1.2 Sparse padding and unconditional size estimates

Assume first only $b \geq 2$. Since $B - 1 > A_*$ and the top coefficient of D_c is 2, we have $D_c(B) > B^{Q_*} > 0$. Also $0 < M_c < N_0$. In fact $S_c > 0$ for every integer g, not merely for $g \geq 1$. Put $v_g = g + 1$ and $A_{\mathrm{sc}} = 2N_0 D_c \geq 2$. Then

$$S_c = A_{\mathrm{sc}} v_g^2 + v_g + N_0 T_* - 1 > 0,$$

because $A_{\mathrm{sc}} v_g^2 + v_g \geq 0$ for every integer v_g, and $N_0 T_* > 1$. Also $T_c > 0$. Consequently, unconditionally in g,

$$R = L^2 + S_c(L-1) + T_c L + L - 1 > L^2.$$

Moreover

$$N^2 = 2^{-20} a^{15} B^{5932} < B^{5947} < B^{11482} < L^2, \qquad b \leq c \leq N.$$

Thus $R \geq N \geq 8$ and $b \leq N, R$ before assuming any carry or square condition. Negative powers here occur only in this comparison proof, never in the equation.

If $1 \leq g < 4bB^W$, integrality gives $1 + g \leq 4bB^W$ and $g < N_0$. The injective coefficients imply

$$|D_c(B)| < \frac{A_* B^{Q_*+1}}{B-1}, \qquad 2(1+g)^2 D_c(B) < \frac{32 A_* b^2}{B-1} B^{K_*+1}.$$

Because $B = 8192 A_* b^2$, the last prefactor is less than $1/128$. Together with $T_* < B^{Q_*+1}/(B-1)$ this gives $0 < J_c < B^{K_*+1}$. Likewise $(B-2)T_* < B^{Q_*+1} \leq B^{K_*+1}$. Consequently

$$0 < S_c, T_c < B^{E_*}, \qquad S_c + T_c < 2B^{E_*} < L.$$

These estimates hold for arbitrary integer $b \geq 2$, not just powers of two.

Now suppose $b = 2^k$. Then $B = 2^\nu$, where $\nu = \kappa + 2k$. The binary mask M_c forces exactly

$$g \,\&\, M_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i}, \qquad 0 \leq n_i < b.$$

Here & is used only in the proof. For such a code, write $h_e = [t^e](1 + \sum n_i t^{w_i})^2 D_c(t)$. Then

$$|h_e| \leq A_*(1 + \textstyle\sum_i n_i)^2 < 1024 A_* b^2 = B/8 < B/4.$$

The unnormalized digits of J_c are $2h_e + \varepsilon_e$, where $\varepsilon_e = 1$ at a target and 0 elsewhere. Starting with incoming carry $c_0 = 0$, normalized carries obey

$$c_{e+1} = \left\lfloor \frac{2h_e + \varepsilon_e + c_e}{B} \right\rfloor, \qquad c_e \in \{-1, 0\}.$$

At a target, its normalized digit avoids $B - 2$ exactly when it is 0 or 1. If $h_e = 0$, the digit is $1 + c_e \in \{0, 1\}$. If $h_e > 0$, it is at least 2; if $h_e < 0$, a borrow normalizes it to a digit at least 2. Thus the target test is equivalent to $h_e = 0$, independently of incoming borrows. There is no need to fill every position with a dense $B/2$ offset. It follows, by splitting at N_0, that

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ codes a common zero of all } p_j.$$

Conversely every common zero has, for arbitrarily large $b = 2^{1+2r}$, such a code with $1 \leq g < bB^W = c$.

1.3 The parity-tuned two-block threshold

Write $L = 2^\lambda$ after $b = 2^k$ has been established. The three disjoint blocks in R give

$$s_2(R) = 1 + s_2(S_c + T_c) + \lambda - s_2(S_c) = \lambda + s_2(T_c) + 1 - \tau_2(S_c, T_c),$$

where s_2 is binary digit sum and $\tau_2(S,T) = s_2(S) + s_2(T) - s_2(S+T) \geq 0$ counts binary carries. The strict inequality $S_c + T_c < L$ is essential. The 31 allowed low positions and the 22 target positions give

$$s_2(T_c) = (W + 23)\nu - 31k - 22, \qquad \lambda = 1 + k + E_*\nu.$$

Therefore

$$\begin{aligned}\lambda + s_2(T_c) + 1 &= 5947\kappa + 11864k - 20 \\ &= 2\big(15\kappa/2 - 10 + 2966(\kappa + 2k)\big) = \log_2 N^2.\end{aligned}$$

Legendre's formula yields $v_2\binom{2R}{R} = s_2(R)$, hence

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

The small factor $2b$ in L calibrates the threshold to a polynomial square without an extra variable, an extra target mask, or a cubic radix.

1.4 An exact polynomial CRT reparameterization

The inherited ratio interface uses $H_{\text{old}} = B_1 + j_{\text{old}}C$ and $JF \mid QF + J(H_{\text{old}} - C)$. Since $E = iJDC^2$,

$$F \equiv 1 \pmod{J}, \qquad F \equiv 1 \pmod{C},$$

and hence $\gcd(J, F) = 1$. The divisibility is equivalent to $J \mid Q$ and $F \mid H_{\text{old}} - C$. Our changed register

$$H = C + F(B_1 + jC)$$

automatically satisfies the latter. It also has the original form, with

$$j_{\text{old}} = 1 + (A^2 - 1)i^2J^2D^2C^3B_1 + Fj,$$

which is a polynomial integer, since $F - 1 = (A^2 - 1)i^2J^2D^2C^4$. This identity holds for arbitrary signed inputs, including $C = 0$.

Conversely, at the canonical positive ratio witnesses, $0 < B_1 < C$ and $j_{\text{old}} \geq 1$. Therefore $z = (H_{\text{old}} - C)/F > 0$ and $z \equiv B_1 \pmod{C}$, so $z = B_1 + jC$ with $j \geq 0$. Thus every canonical witness is represented. No extra unknown or cofinal restriction is introduced. The merger now has $u = J$ rather than JF. This CRT step is inherited from the separately identified sparse-Pell packet, not claimed as an original theorem here.

1.5 The interval guard and the Jones interface

If $\Delta > 0$ and $b \geq 2$, then $K \neq 0$ and $0 < g < c$. The identity $4g(c-g) = c^2 - (c-2g)^2 \leq c^2$ implies $4\delta^2 < K^2$. Along with $s \geq 1$, $\phi \geq 0$, and $w \neq 0$, this gives $C, Y > 0$ and $K > 0$. The usual square-gap argument forces $h \geq 1$: if $h \leq 0$, then $1 \leq K \leq R + 1 < P$; $K = 1$ would force $P - 1 \mid R$, which is impossible, while for $2 \leq K < P$,

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2.$$

Thus a_0 cannot be a square in that branch.

For precision, instantiate the free ceiling in `Jones1982.Nine.ratio_sound` by $G_{\mathrm{aux}} = 8gK^2+1$, rather than assuming its old margin condition. The positive integer $K^2 - 4\delta^2$ then gives

$$(K^2 - 4\delta^2)G_{\mathrm{aux}} - 8gK^2 > 0.$$

This ceiling is a proof-side abbreviation; the ratio registers do not otherwise depend on it.

Replace i by $|i|$, which changes no condition. To handle signed j, $u \mid t$ now means $J \mid Q$, while $F \mid H - C$ holds identically. Also $F \equiv 1 \pmod{D}$, $G \equiv 1 \pmod{D}$, and $G \equiv A \pmod{F}$ show that the positive integers D, F, I are pairwise coprime. Hence $a_1 = DFI$ square makes each a square. Put $d = G^2 - 1 > 0$ and $x = \sqrt{I}$. The norm-one unit $x + H\sqrt{d}$ is positive even if $H < 0$. A positive power of $G + \sqrt{d}$ is the identity modulo CF; multiplying by sufficiently large such powers gives a Pell solution $H' > B_1$ with $H' \equiv H \pmod{CF}$. Then $j' = (H' - B_1)/C > 0$, all square conditions remain true, and $J \mid Q$ remains true. The literal positive Jones interface therefore applies and gives the power-of-two base and central-binomial condition required above.

For completeness, take any common source zero, choose $b = 2^{1+2r}$ large enough, and set $\epsilon = 2^r$. The direct source code has $32g < G_c$, so the literal ratio completeness theorem supplies its canonical witnesses. In `Ratio.lean`, `exists_B25core` sets

$$\omega = 2^{2R+1-k}, \qquad U = (N^2/b)2^{2R+1}.$$

Since $k = 1 + 2r$ is odd and $k < 2R + 1$, this ω is a positive square. No claim that the canonical s is a square is needed.

The same canonical construction has $\rho_0 = (U + 1)^{2R}/U^R = Y + f$ with $0 \le f < 2^{2R}/U = b/(2N^2)$ and $Y \ge 1$. The source's `ratio_core` gives

$$|C/K - \rho_0| \le \frac{2(Y + f)}{Y(U + 1)} < \frac{4}{U}, \qquad |C/K - Y| < \frac{b}{N^2}.$$

The last inequality uses $R \ge 8$. Since $c, b \le N$ and $g(c - g) \ge 1$, it follows that

$$\frac{c^2(C - KY)^2}{K^2} < \frac{c^2b^2}{N^4} \le 1 \le g(c - g),$$

so the very same canonical witnesses, translated by the exact CRT formula, satisfy $\Delta > 0$. This is a cofinal square-base proof, not an unsupported restriction of an existing slack variable to squares.

1.6 Raw two-root recovery, including the zero spectator branch

For every integer assignment, $u = J = 4A - 5 \neq 0$. The polynomial

$$\Pi(x) = (x^3 - x - 1)(x^2 + 23)$$

has no rational root and has a root modulo every positive integer. For an odd prime other than 23, either the cubic has a simple root or it is irreducible, whose Frobenius three-cycle makes its nonzero discriminant -23 a square. The quadratic then has a simple root. At 23 the cubic has the simple root 3; at 2, $-23 \equiv 1 \pmod{8}$ admits compatible square roots. Hensel lifting and CRT finish the assertion. Thus $\Theta = u^5\Pi(t/u + r) \neq 0$ for every assignment, and

$$\exists r \quad Zu^5 \mid \Theta \quad \Longleftrightarrow \quad Z \neq 0 \text{ and } u \mid t.$$

Indeed a root modulo $|Z|$ proves the reverse implication once $u \mid t$. If $v_p(u) = \alpha > \beta = v_p(t)$, the two factors of Θ have valuations 3β and 2β, contrary to $u^5 \mid \Theta$. If $Z = 0$, divisibility would require $\Theta = 0$, already impossible.

At a zero of the equation, $\Sigma = 0$ is impossible because then the polynomial is $\Theta^4 \neq 0$. In particular $\epsilon, w, i, z_1, z_2, z_3, a_0$ are nonzero. Every $d_j \equiv 2 \pmod{3}$ is nonzero, and every $B_j^* \equiv 1 \pmod{4}$ is nonzero. Thus C_2, C_* are nonzero.

No coincident radicands. One has $C_* \neq a_1$ without any positivity assumption. If $a_1 = 0$ this follows from $C_* \neq 0$. Otherwise $B_3^* \neq 1$, and it cannot be -1 because it is 1 modulo 4. But

$$\gcd(B_3^*, a_1) = 1,$$

while $C_* = a_1$ would make B_3^* divide a_1. This would force $B_3^* = \pm 1$, a contradiction.

Put $q = X/\Sigma$. The raw norm equation implies

$$(q^2 + C_* - a_1)^2 = 4C_* q^2.$$

If $q = 0$, the norm would give $C_* = a_1$, just excluded. Therefore

$$r_0 = \frac{q^2 + C_* - a_1}{2q}, \qquad r_1 = q - r_0$$

are rational, with $r_0^2 = C_*$ and $r_1^2 = a_1$. Rational square roots of integers are integers. Hence C_*, a_1 are integer squares, $q \in \mathbb{Z}$, and $\Sigma \mid \Theta$. In particular $J \mid Q$. No complex-root separation assumption is used.

Recover the first two guards before ruling out $a_1 = 0$. The factors a_0, B_1^*, B_2^*, B_3^* are pairwise coprime, because each new factor is 1 modulo its entire preceding product. Their nonzero product C_* is a square. Thus each absolute factor is a square. Since $B_j^* \equiv 1 \pmod 4$, every B_j^* is positive: a negative square cannot be 1 modulo 4. The product sign then forces $a_0 > 0$, also a square. All prefixes are now positive, so $d_1, d_2 > 0$, yielding $s \geq 1$ and $\phi \geq 0$.

At this point the last factor might still equal 1 if $a_1 = 0$; this branch must not be silently discarded. The unconditional source estimate above has already proved $R > L^2 > 0$ for *every* signed g. Since $\epsilon, w \neq 0$ and $s \geq 1$, we have $b \geq 2$, $N \geq 8$, $U, Y > 0$, $A \geq 2$, and, using $\phi \geq 0$, $C > 0$. Then

$$D \geq A^2 > 0, \qquad F \geq 1, \qquad CD + 1 - A > 0, \qquad G \geq A > 1, \qquad I \geq 1.$$

Consequently $a_1 = DFI > 0$. Now $B_3^* \neq 1$, and its positivity with $C_2 a_1 z_3^2 > 0$ forces $d_3 > 0$, hence $\Delta > 0$. This closes the zero-spectator branch without adding a_1 to the nonzero multiplier. The interval and ratio arguments above therefore apply.

Completeness. Start from the canonical positive ratio witnesses, translated by the CRT parameterization. They have a_0, a_1 positive squares, $J \mid Q$, and all three strict sign conditions. For $j = 1, 2$, $d_j C_{j-1}$ is positive nonsquare because $d_j \equiv 2 \pmod 3$ and C_{j-1} is a square. A nontrivial Pell solution $x^2 - d_j C_{j-1} y^2 = 1$ can be chosen with y even, so $z_j = y/2 \neq 0$ makes $B_j^* = x^2$. For the last guard, $d_3 C_2 a_1$ is likewise positive nonsquare, since $C_2 a_1$ is a positive square. The same construction gives B_3^* square and C_* square. The intersective lemma chooses r with $\Sigma \mid \Theta$. Finally choose

$$v = \Theta/\Sigma + \sqrt{C_*} + \sqrt{a_1} \in \mathbb{Z}.$$

Then the raw norm vanishes. Both directions are proved. By the inherited compiler theorem, integer-zero existence is equivalent to a finite ZFC refutation, and ZFC proves its negation equivalent to Con(ZFC).

1.7 Exact degree, comparison, and verification scope

The exact relevant register degrees are

	a_0	a_1	C_2	C_*	Σ	X
deg	652210	6592598	2608849	12185330	1119171	1119172

and $\deg A_0 = 14423672$, $\deg A_1 = 8830940$. The degree of A_0 strictly exceeds those of X^2 and A_1. The output has leading term A_0^2. Its diagonal leading coefficient is

$$72338\, a^{7211252}\eta^{960} \pmod{1000003},$$

which is nonzero since a and η are powers of two. The circuit upper bound and this specialization together prove exact degree 28 847 344.

At the identical fixed source and selective code, the previous seeded construction has degree 57 866 184 and 759 operations. The coprime spectator alone gives degree 46 680 720 and 758 operations; combining it with CRT gives degree 28 847 344 and 754 operations, a 50.15% degree reduction against the seeded predecessor. All have thirteen integer unknowns. This is not a twelve-variable construction, a variable-count record, a global optimum, or a claim that coefficient storage decreased.

The canonical circuit additionally factors the merger cubic as $\rho(\rho^2 - u^2) - u^3$, saving one multiplication without changing the polynomial. The metric counts every live binary addition, subtraction and multiplication, including scalar multiplication and fixed exponentiation. Reuse and fixed coefficient leaves are free. The code independently replays literal formulas over three finite fields, checks signed and zero branches, the norm identity, the CRT identity, nonvacuous Pell witness examples, source positivity, sparse carries, and the inherited fixed-byte hashes. Those are finite implementation audits, not a replacement for the universal proof or a new kernel-checked formal theorem.

Run `python build_jones.py`, `python check_jones.py`, and `python -O check_jones.py`. The complete equation and fixed binding are in the emitted DAG and `FIXED_BINDING_OLD_SOURCE.json`. Source provenance is retained in `SOURCE_PINS.json`; the CRT source packet and its archive hash are recorded separately in `CRT_PROVENANCE.json`.